

Ahnaf Shahriar

[Email](#) | [LinkedIn](#) | [Github](#) |

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Computer Engineering

Sept. 2021 – Apr. 2026

- Recipient of Richard & Elizabeth Madter Entrance Scholarship and President's Scholarship of Distinction
- **Relevant Courses:** Algorithms and Data Structures II, ML for chip Design, Distributed Computing, Compilers, Real-Time Operating Systems

EXPERIENCE

Undergraduate Research Assistant

May. 2025 – Present

University of Waterloo

Waterloo, ON

- **Research:** Conducting Case study on Static Analysis type checking for Java.
- **Optimization:** Implementing bytecode storage on NULLness checker for EISOP framework.

Wireless Firmware Developer

Jan. 2025 – Apr. 2025

ON Semiconductor Corporation

Waterloo, ON

- **SDK Development:** Integrated *I3C support* and *BLE* profile services into sample code.
- **Wireless Connectivity:** Developed and integrated cutting-edge *LE Audio*, *HFP*, and *A2DP* functionality into an all-in-one Application.
- **Wireless Manager:** Modularized and implemented a state machine for power efficiency by *25%*
- **Test Automation:** Automated SDK test cases, reducing manual testing efforts by up to *80%*

IC Design and Verification Intern

May 2023 – Aug. 2023, Jan. 2024 – May 2024

NXP Semiconductors Canada

Kanata, ON

- **IP Design:** Designed multiple IP blocks including *proprietary ECC IP* for dataplane processing chips.
- **Timing Analysis:** Spearheaded critical path improvements in IP block to increase Clock Frequency by *50%*.
- **Functional Testing:** Designed brand new End-to-End functional tests in simulating *10GBps+* traffic for IP.
- **Unit Test Planning:** Created Simulation scenarios for testing High speed Dataplane Processing features.

Embedded Software Engineering Intern

Sept. 2022 – Dec. 2022

Synapse Product Development

Seattle, WA

- **Prototyping:** Leveraged Zephyr RTOS to create a proof of concept on *NRF52 BLE* device.
- **Python APIs:** Developed company specific lab automation software for equipment from *Agilent*, *Keysight*, *NI*, *Tektronik*.
- **Automation:** Streamlined testing and in house procedures using *Python* and *Bash*.
- **Driver Development:** Designed and implemented *drivers* of automated PCB testing Device(*I2C*, *UART*)

PROJECTS

VHDL Compiler | *Java*

- Performed **Syntactic and Semantic** analysis in a layered approach.
- Generated diagrams for circuits in **NAND and CNF** form, with optional **waveform simulation** using bitstreams.
- Implemented modular **x86/ARM ASM and Java** code generation and simulation layers to support cross-platform execution and optimization.

CubeSolver | *C++, Linux, NCurses, OpenGL*

- A Program that can solve **any Rubix Cube** you scramble. Optimized for bitwise operations.
- Implemented optimized **Algorithms** for solving the cube in **less than 20 moves**.
- Optional 3D Visualization using **OpenGL** for movements.

TECHNICAL SKILLS

Languages: C/C++, Python, Rust, Java, Javascript, SQL, Bash and shell scripting, ASM

Frameworks: LLVM, OpenGL, ROCm, Node Js, ElectronJS, Apache Spark, Hadoop

Developer Tools: Git, Docker, Kafka, Qemu, Matlab, Jupyter Notebook

Libraries: pandas, NumPy, Sci-kit, Matplotlib, pyTorch